

Aleksandr Kowalski

+44 20 7946 0958 | Email: aleksandr.kowalski@gmail.com | LinkedIn: [aleksandr-kowalski](https://www.linkedin.com/in/aleksandr-kowalski) | GitHub: [AleksandrKowalski](https://github.com/AleksandrKowalski) | <https://aleksandr.dev>

EDUCATION

ETH Zurich (Swiss Federal Institute of Technology) | *MSc Computer Science* 2010–2012

- Thesis on distributed systems optimization achieved 9.2/10 (top 3% cohort); published 2 peer-reviewed papers on lock-free concurrent data structures in IEEE Transactions on Parallel and Distributed Systems
- Led Systems Engineering Club with 150+ members; organized 8 technical workshops on concurrency patterns, attracting speakers from Google and Microsoft

University of Warsaw | *BSc Mathematics & Computer Science* 2006–2010

- Graduated summa cum laude with 3.95/4.0 GPA; awarded Faculty Excellence Award for contributions to algorithmic research
- Conducted undergraduate research on randomized algorithms under Prof. Marek Cygan; co-authored 1 publication in competitive programming domain

EXPERIENCE

Stripe | *Principal Engineer, Infrastructure & Performance* 2019–Present

- Architected and led 12-person team delivering next-generation payment processing pipeline; reduced P99 latency by 340ms (from 450ms → 110ms) and increased throughput 2.8x (180K → 504K req/sec) serving \$2.2T+ annual payment volume
- Designed distributed consensus protocol for real-time settlement reconciliation across 18 global regions; achieved 99.9997% consistency (3 nines improvement) while cutting operational overhead by \$4.2M annually
- Mentored 5 senior engineers to principal-track roles; established incident command training program reducing MTTR by 62% and incident severity by 58% across 200+ microservices

Cloudflare | *Senior Staff Engineer, Network Systems* 2016–2019

- Built HTTP/3 and QUIC optimization layer deployed across 280+ edge locations processing 40M req/sec; improved median latency 230ms → 85ms (63% reduction) and reduced packet loss recovery overhead by 45%
- Led platform reliability initiative improving global DNS resolution SLA from 99.95% → 99.999% (4-nines improvement); saved customers ~\$18M in potential downtime costs annually
- Championed adoption of eBPF-based network telemetry across infrastructure; cut observability costs by 67% (\$2.8M savings) while increasing metric resolution from 60s to 1s intervals

Shopify | *Staff Engineer, Platform & Databases* 2013–2016

- Redesigned distributed sharding architecture for primary transactional database (900GB+, 250K queries/sec); achieved 4x improvement in query latency (from 280ms → 70ms) and enabled 5.2x capacity scaling without hardware upgrades
- Implemented real-time replication system for cross-datacenter failover; reduced RTO from 45 minutes → 4 minutes and RPO from 30s → <100ms, earning \$6.8M in SLA credit avoidance
- Managed platform infrastructure for 850K+ merchant stores; optimized resource allocation reducing cloud spend by \$3.1M YoY while maintaining 99.98% uptime SLA across containerized workloads

Google | *Senior Software Engineer, Infrastructure* 2011–2013

- Developed high-performance RPC framework for MapReduce clusters processing 8 exabytes/day; cut request serialization overhead by 73% through custom binary protocol and reduced tail latency (p99) from 320ms → 80ms
- Owned cluster resource scheduler serving 14K compute nodes across 6 datacenters; improved resource utilization by 31% (\$12M annual savings) and reduced job startup time by 52%
- Co-authored 2 internal papers on distributed tracing and adaptive batching; knowledge transferred to 28 engineers across 4 teams reducing incident resolution time by 44%

PROJECTS

CacheMeld | *Rust, C++, Redis, Memcached* 2020–Present

- Open-source distributed cache coherence framework with intelligent multi-tier invalidation; 4,200+ GitHub stars, 187K monthly npm downloads, and adopted by 40+ companies including 2 Fortune 500s
- Achieved 95th percentile cache hit rate 87.4% on real-world workloads vs. 71% for competitors; provided 4.2x memory efficiency through adaptive compression, reducing production cache footprint by \$800K annually for users

VectorFlow | *Go, Python, TensorFlow, Protocol Buffers* 2018–2022

- Machine learning inference serving platform with dynamic batching; achieved 8,600+ GitHub stars and integrated into 12 production systems at Stripe, Shopify, and Figma
- Delivered 6.8x throughput improvement (from 420 → 2,856 inferences/sec per GPU) through model quantization and batch optimization; reduced inference latency from 142ms → 38ms while cutting GPU costs by 58%

RaftControl | *Go, Protobuf, etcd, Kubernetes* 2019–2021

- Consensus algorithm testing and verification framework; 2,100+ GitHub stars with adoption in 18 production systems; featured in 3 academic papers on consensus correctness
- Reduced consensus failure detection time from 14s → 1.2s through improved leader election logic; generated 500K+ test scenarios covering edge cases, catching 47 previously unknown bugs in popular implementations

EXTRACURRICULAR

Strange Loop Conference & QCon | *Technical Speaker & Program Committee Member* 2017–Present

- Delivered 6 keynotes and technical talks on distributed systems, concurrency, and performance optimization with combined audience reach of 8,400+ engineers; talks achieved 4.7/5.0 average speaker rating
- Served on program committee for 4 conferences selecting 180+ talks from 2,100+ submissions; mentored 12 first-time speakers resulting in 9 accepted presentations and fostering emerging voices in systems community

Open Source Systems Mentorship Initiative | *Principal Mentor & Advisor* 2016–Present

- Actively mentor 7 open-source projects across Rust ecosystem; 4 mentees promoted to core maintainer roles within 18 months; mentored codebase now maintains 99.2% test coverage with zero critical CVEs in 3-year period
- Authored comprehensive distributed systems and performance optimization guide (100K+ GitHub clones, 340K readers); established weekly office hours attended by 45+ engineers, providing technical guidance on architecture decisions and code reviews

TECHNICAL SKILLS

Languages – Go, Rust, C++, Python, Java, SQL, Bash, JavaScript

Technologies – Kubernetes, Distributed Systems, Consensus Algorithms (Raft, Paxos), Protocol Buffers, gRPC, REST APIs, Event Streaming (Kafka), PostgreSQL, DynamoDB, Redis, Message Queues (RabbitMQ), TensorFlow, Docker, Terraform, etcd, Datadog, Prometheus

Tools – Git, Linux, Docker, Kubernetes, Terraform, CloudFormation, GitHub Actions, Jenkins, Prometheus, Grafana, Datadog, PagerDuty, VirtualBox, GCP, AWS, Azure, Bazel